

Problem 1 - Washing Machine Mount Design

The Matlab code for this problem is listed in Appendix 1.

Problem 1a

Equation 1 calculates the force produced by the imbalanced 1kg mass at a 0.5 m radius from the center of the washing machine as a function of angular velocity,

$$F = m \cdot \omega^2 \cdot r \quad (1)$$

where m is 0.5 kg, ω is the angular velocity in $\frac{\text{radians}}{\text{s}}$, and r is 0.5 m.

Figure 1 shows the force produced by the 1 kg load as a function of angular velocity in RPM.

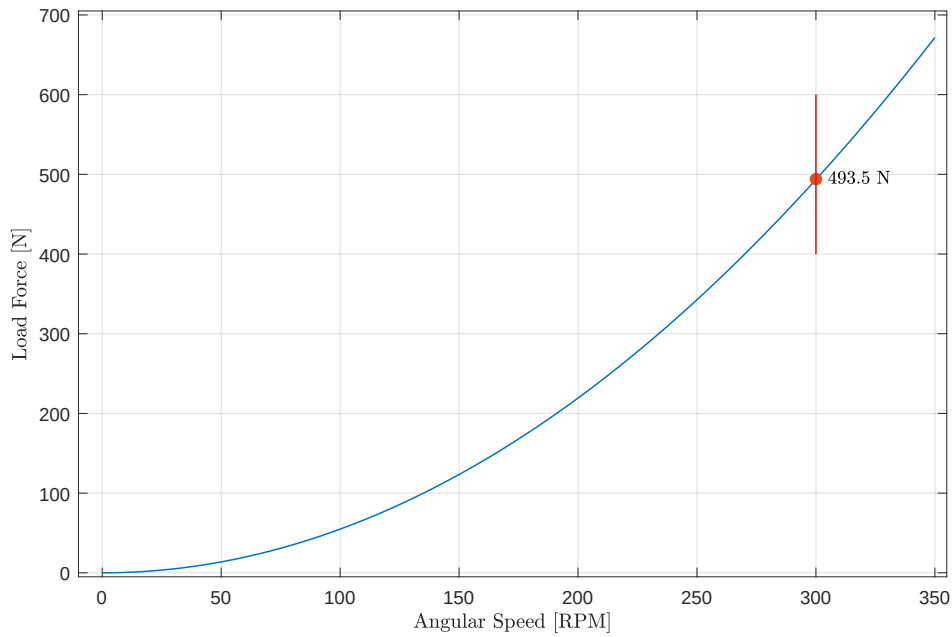


Figure 1: Load force versus washing machine angular speed.

At 300 RPM, or 5 Hz, the imbalanced load force amplitude is 493.5 N.

Problem 1b

The required spring stiffness, k_s , is $9,810 \frac{\text{N}}{\text{m}}$. The mount's natural frequency, w_o , is 1.5 Hz.

Problem 1c

Figure 2 shows the transmission loss of the system for a set of damping ratios ranging from 0.001 to 1.

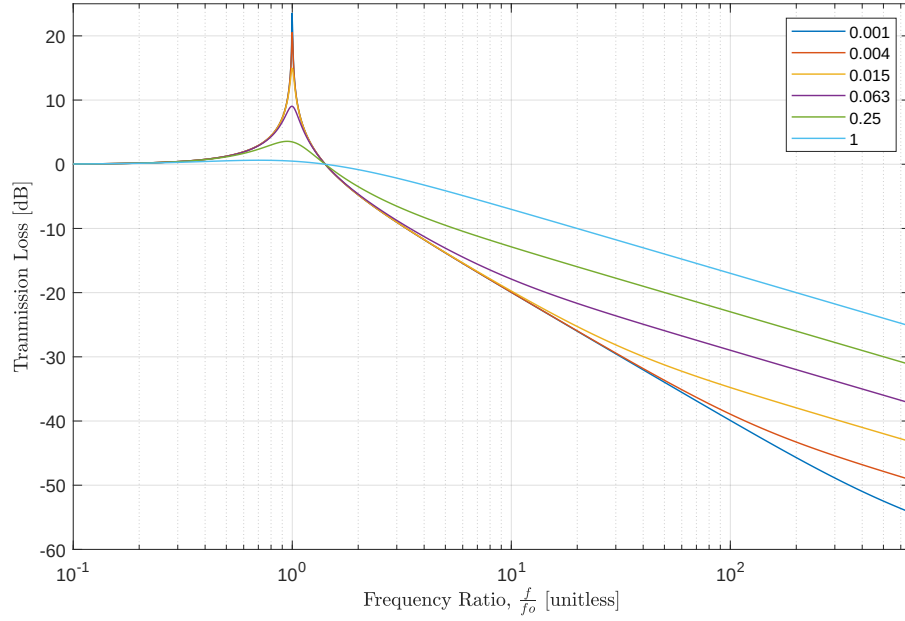


Figure 2: Transmission loss as a function of damping ratio. The natural frequency, f_o , is 1.5 Hz.

Problem 1d

Figure 3 shows the force applied to the foundation.

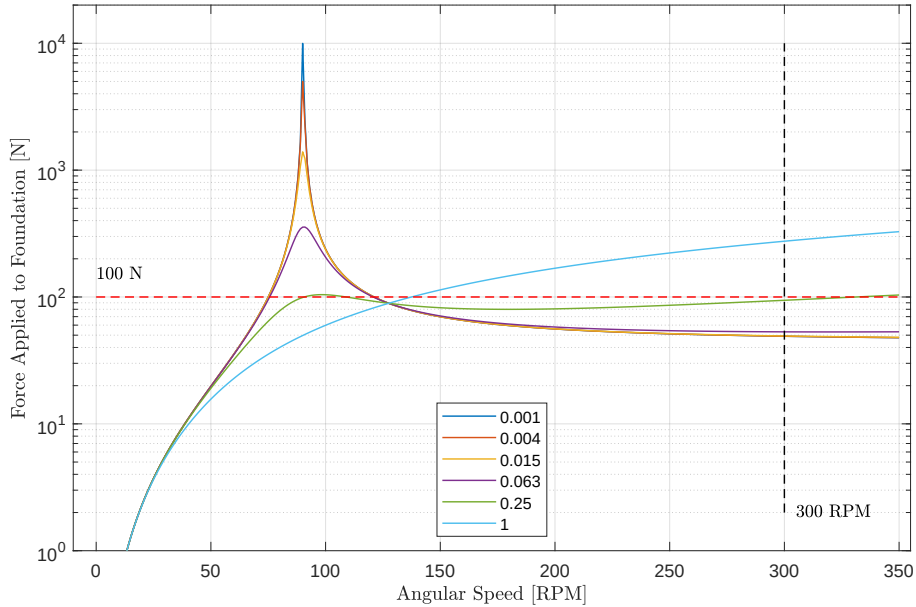


Figure 3: Force applied to the foundation as a function of damping ratio.

No damping ratio makes the applied force less than 100 N across the 0 to 350 RPM operating range.

The best damping ratio is $\zeta = 0.25$. The force applied to the foundation still exceeds the 100 N target, but in only two small RPM regions: 91 to 109, and 331 to 350 RPM. In both of these regions, the maximum force is approximately 104 N.

Problem 1e

The viscous damping coefficient is $519.4 \frac{\text{kg}}{\text{s}}$ based on a total mass of 110 kg.

Figure 4 shows the admittance.

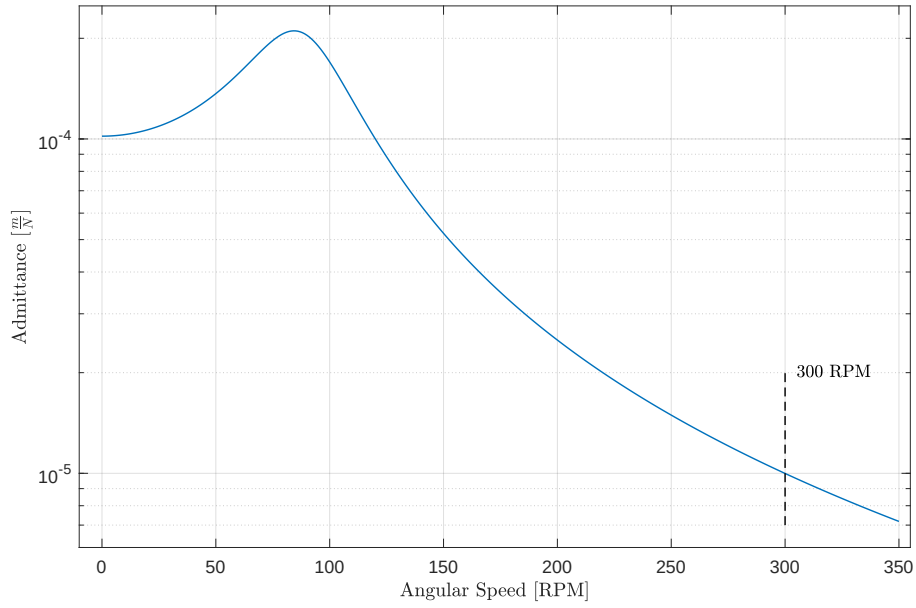


Figure 4: Admittance versus RPM.

The admittance at 300 RPM is $10^{-5} \frac{\text{m}}{\text{N}}$.

Problem 1f

Figure 5 shows the displacement of the washing machine.

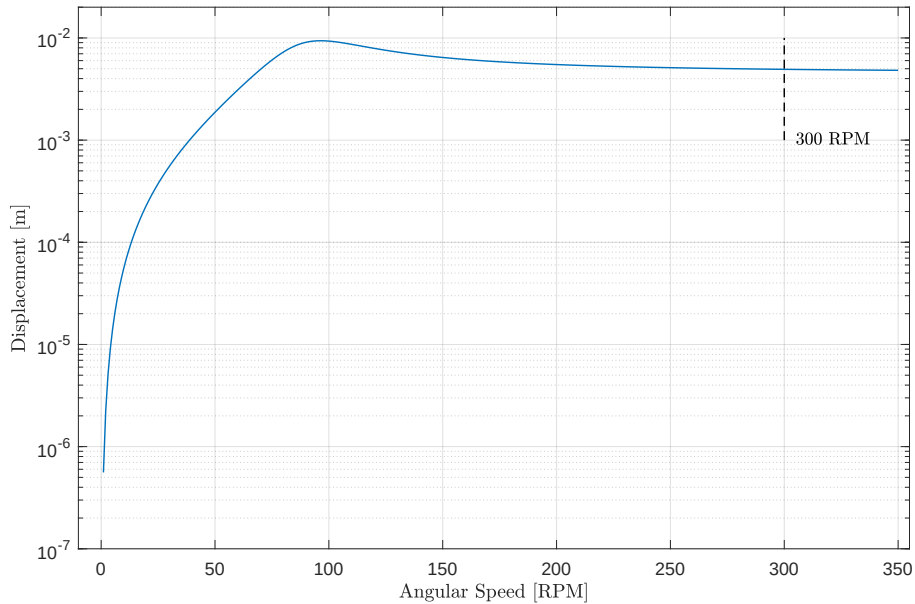


Figure 5: Displacement versus RPM.

The displacement of the washing machine at 300 RPM is approximately 4.9×10^{-3} m. The washing machine undergoes larger momentary displacements as it spins up to 300 RPM, in particular around 100 RPM.

The maximum displacement is 9.4×10^{-3} m, which is very large for prolonged operation.

Problem 2 - Washing Machine Mount Design (2 DOF)

The Matlab code for this problem is listed in Appendix 2.

Problem 2a

Figure 1 illustrates the two-stage vibration mount showing the washing machine and the “raft”.

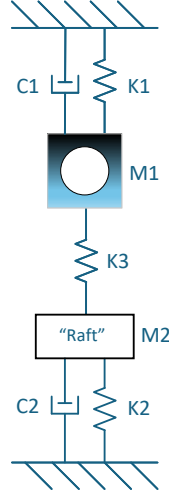


Figure 1: Two-stage vibration mount system for the washing machine.

Problem 2b

Table 1 lists the initial stiffness and damping coefficients for the system.

M1	110	kg
K1	100	$\frac{\text{N}}{\text{m}}$
C1	5	$\frac{\text{kg}}{\text{s}}$ ¹

M2	100	kg
K2	100	$\frac{\text{N}}{\text{m}}$
C2	5	$\frac{\text{kg}}{\text{s}}$ ²

M3	0	kg
K3	1,000	$\frac{\text{N}}{\text{m}}$
C3	0	$\frac{\text{kg}}{\text{s}}$ ³

Table 1: Initial values of stiffness and damping coefficients.

Damping was implemented using the damping vector⁴.

¹Element of damping vector.

²Element of damping vector.

³Element of damping vector.

⁴Not equivalent to using a complex-valued stiffness vector; the imaginary part is the damping for the respective spring

Problem 2c

Figure 2 shows the admittance of the washing machine and the raft.

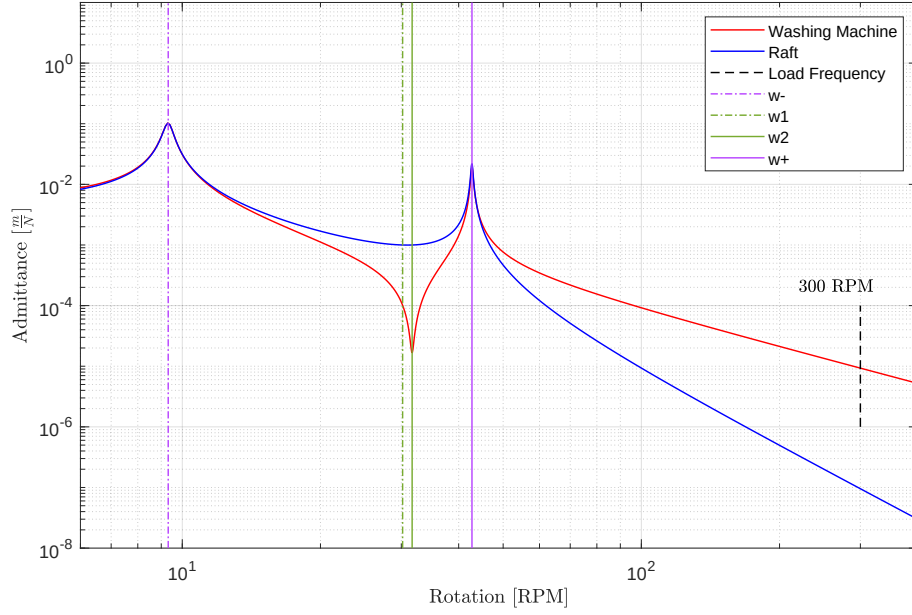


Figure 2: Admittance of the washing machine and the raft with a force on the washing machine.

The admittance of the washing machine is $9.31 \times 10^{-6} \frac{\text{m}}{\text{N}}$ at 300 RPM.

Problem 2d

Figure 3 show the displacement of the washing machine and the raft.

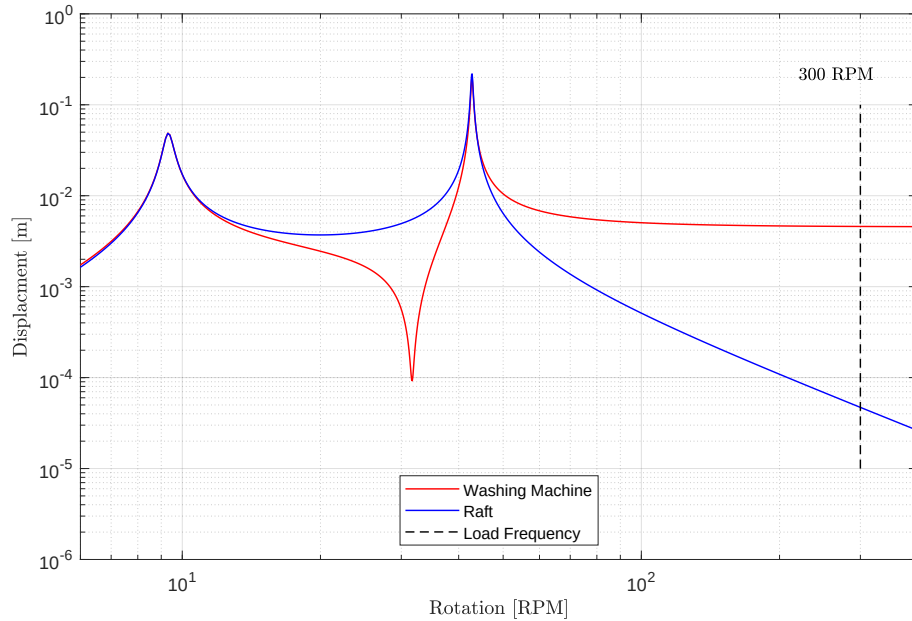


Figure 3: Displacement of the washing machine and the raft with a force on the washing machine.

At 300 RPM, the washing machine displacement is $4.6 \times 10^{-3} \text{ m}$, less than $5.0 \times 10^{-3} \text{ m}$ for the first-order system. The displacement of the raft is $4.7 \times 10^{-5} \text{ m}$, which is 200 times less than the 1 DOF system.

Problem 2e

Figure 4 shows the force applied to the ground by the raft.

The force is calculated by multiplying the displacement of the raft by the K_2 spring constant.

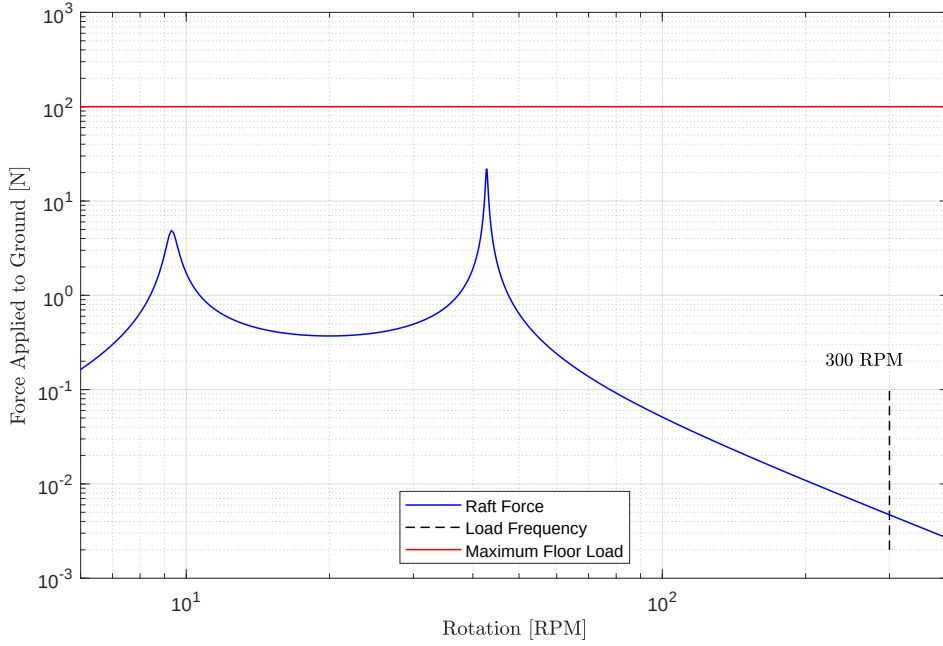


Figure 4: Force applied to the ground by the raft.

The ground force at 300 RPM is 4.7×10^{-3} N. The force on the ground is always less than 100 N.

Problem 2f

The performance of this system was improved by *lowering the stiffness of the coupling spring*, k_3 . Lowering k_3 decreases the separation of the two resonance peaks and shifts them to lower frequencies. This results in lower admittances for the washing machine and the raft at 300 RPM. However, due to the quadratic characteristic of the load force with RPM and the second-order roll-off of the admittance curve for the washing machine, optimization is restricted.

In Figure 3, displacement of the washing machine after the second resonance frequency “plateaus” with much less roll-off relative to the raft displacement. This limits optimization.

Problem 2g

Table 2 compares the admittance, displacement, and ground force for the initial and optimized systems.

	Admittance [$\frac{\text{m}}{\text{N}}$]	Displacement [m]	Ground Force [N]
Initial Design	9.31×10^{-3}	4.59×10^{-3}	4.7×10^{-3}
Optimized Design	9.22×10^{-6}	4.55×10^{-3}	4.62×10^{-5}

Table 2: Performance comparison of the initial and optimized designs.

It is apparent that the optimized design greatly reduces the admittance and the ground force. However, the change in displacement is small in comparison.

Problem 3 - Washing Machine Dynamic Vibration Absorber Design

The Matlab code for this problem is listed in Appendix 3.

Part 3a

Figure 1 illustrates the dynamic vibration absorber (DVA) system.

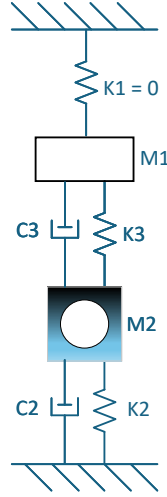


Figure 1: Dynamic vibration absorber system for the washing machine.

Table 1 lists the stiffness and damping coefficients for the system.

M1	15	kg
K1	0	$\frac{\text{N}}{\text{m}}$
C1	0	$\frac{\text{kg}}{\text{s}}$

M2	110	kg
K2	9,810	$\frac{\text{N}}{\text{m}}$
C2	1,962	$\frac{\text{kg}}{\text{s}}$ ¹

M3	0	kg
K3	1e5	$\frac{\text{N}}{\text{m}}$
C3	1e4	$\frac{\text{kg}}{\text{s}}$ ²

Table 1: Stiffness and damping coefficients for the DVA system.

Damping was implemented by adding with an imaginary component to the respective spring stiffness³.

¹0.2 fraction of respective spring stiffness.

²0.1 fraction of respective spring stiffness.

³Not equivalent to using a damping vector in the argument to the given Matlab nDOF function script-file.

Part 3b

Figure 2 shows the admittance response of the original system and DVA system based on the equations presented in the tutorial lecture on DVA system analysis.

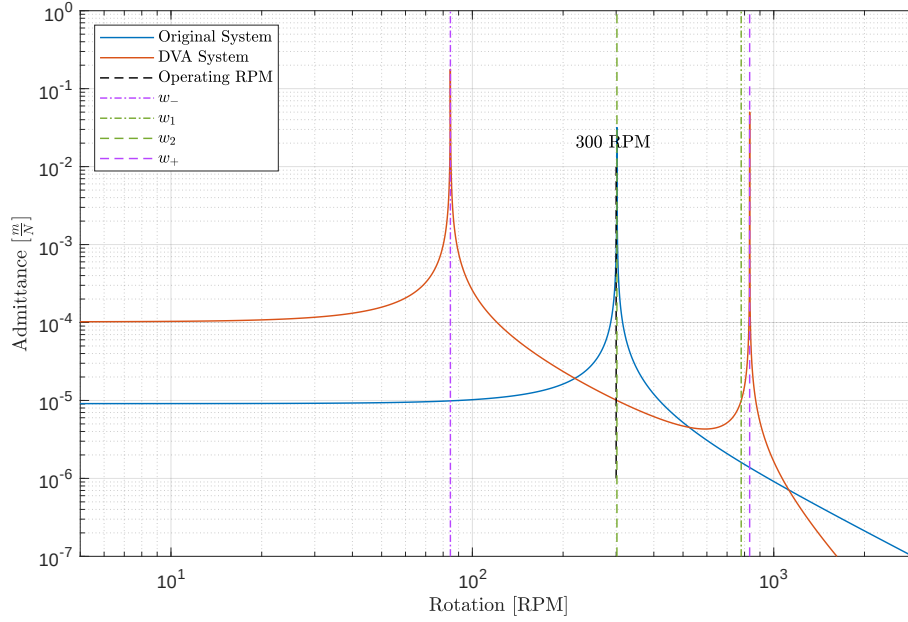


Figure 2: Admittance responses based on the equations from the DVA tutorial lecture.

The washing machine admittance at 300 RPM is reduced from the blue curve resonance peak to a point between the two resonances on the red curve.

Part 3c - Admittance

Figure 3 shows the admittance response of the DVA system using the nDOF function script-file. The stiffness of the coupling spring, k_3 , is set to place the null of the admittance response at the 300 RPM operating state of the washing machine. *The admittance at 300 RPM is approximately $1.2 \times 10^{-6} \frac{\text{m}}{\text{N}}$* ⁴

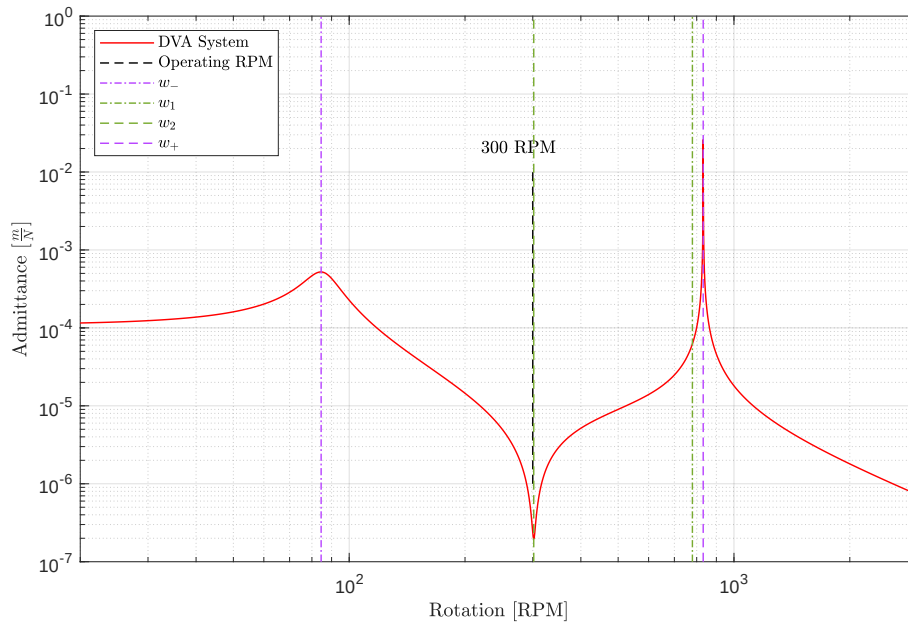


Figure 3: Solution using nDOF script-file function.

⁴This is theoretical and not realistic in practical terms, but the admittance is reduced.

Part 3d - Displacement

Figure 4 shows the displacement of the washing machine as a function of RPM. The ramp up to 300 RPM occurs quickly so that the large displacements are transient and the maximum speed is 350 RPM (the large displacement associated with the resonance peak at higher frequencies is not reached).

The displacement at 300 RPM is approximately 38.2×10^{-6} m.

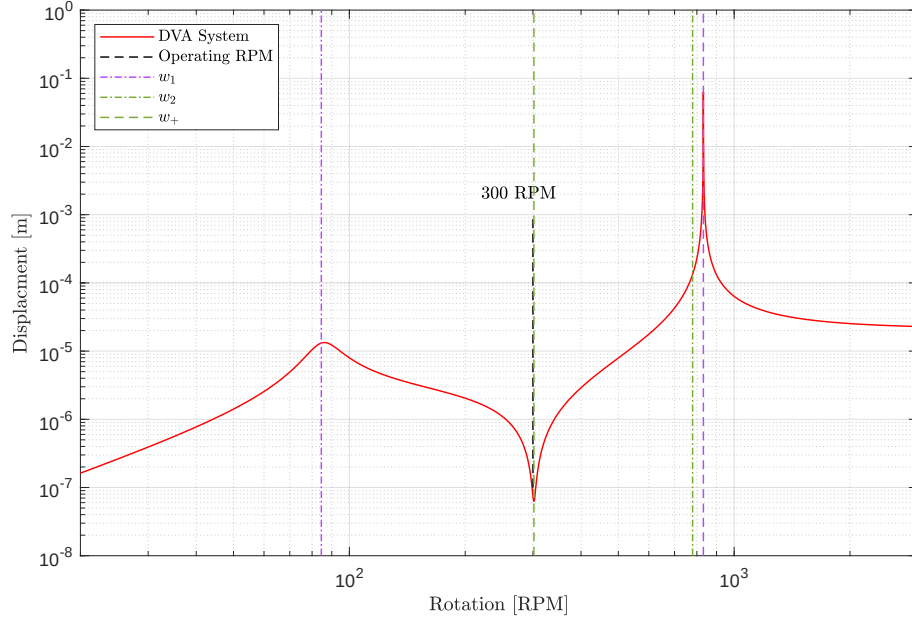


Figure 4: Displacement of the washing machine.

Part 3e

Figure 5 shows the force applied to the floor by the washing machine.

The force applied to the floor at 300 RPM is approximately 3.75×10^{-3} N.

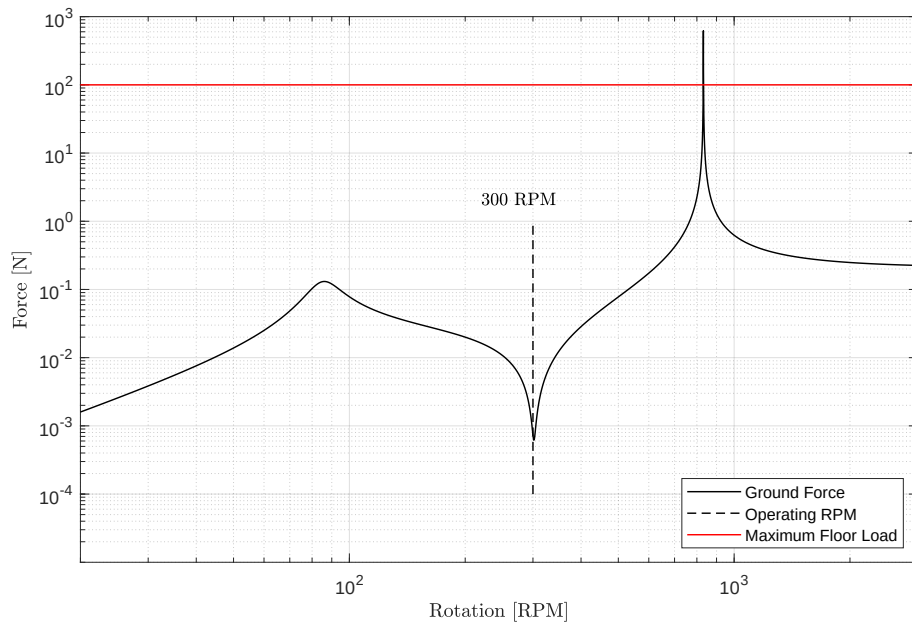


Figure 5: Force applied to the floor.

Problem 4 - Washing Machine Abuse Testing

The Matlab code for this problem is listed in Appendix 4.

Part 4a

Equation 2 is the impulse response for a 1 DOF system.

$$x(t) = \frac{1}{m \cdot w_d} \cdot e^{(-w_o \cdot \zeta \cdot t)} \cdot \sin(w_d \cdot t) \quad (2)$$

where m is the mass of the washing machine, w_d is the damped natural frequency, w_o is the natural frequency, and ζ is the damping ratio.

Figure 1 shows plots of a 1 kg impulse force and the corresponding response of the washing machine.

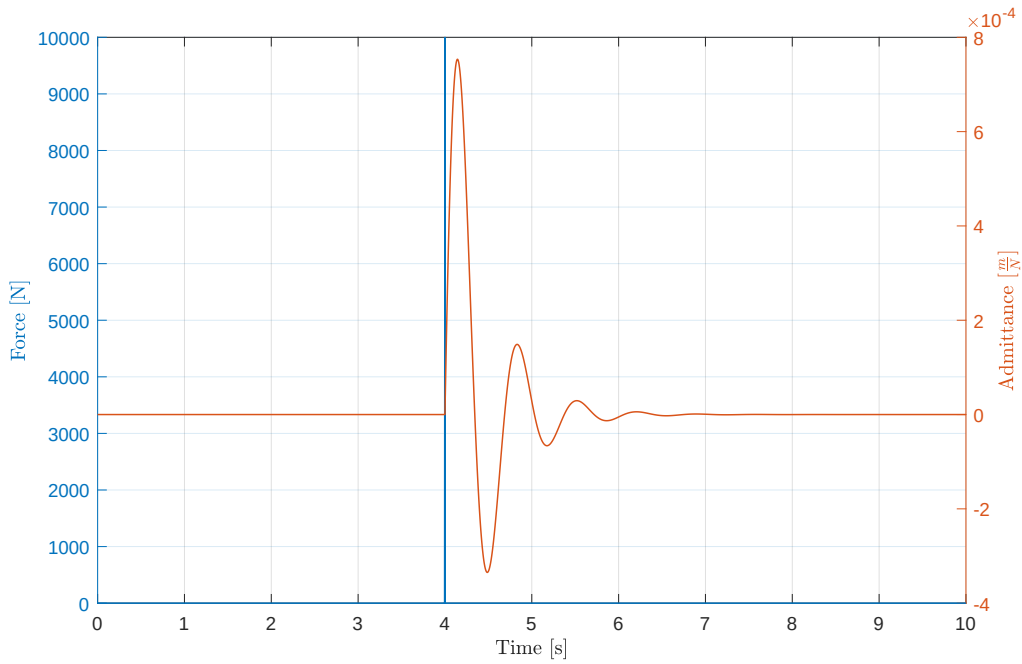


Figure 1: Washing machine impulse response. w_o is $9.4 \frac{\text{radians}}{\text{s}}$, w_d is $9.1 \frac{\text{radians}}{\text{s}}$, and ζ is 0.25.

Part 4b

A operational sequence for the washing machine created using a sinusoidal forcing function and the impulsive loading with the 5 kg brick. The sequence is as follows,

1. Washing machine is off.
2. Sinusoidal loading starts at 1 second.
3. 5 kg brick is inserted at 4 seconds.
4. Sinusoidal loading ends at 8 seconds.

Figure 2 shows the response sequence of the washing machine.

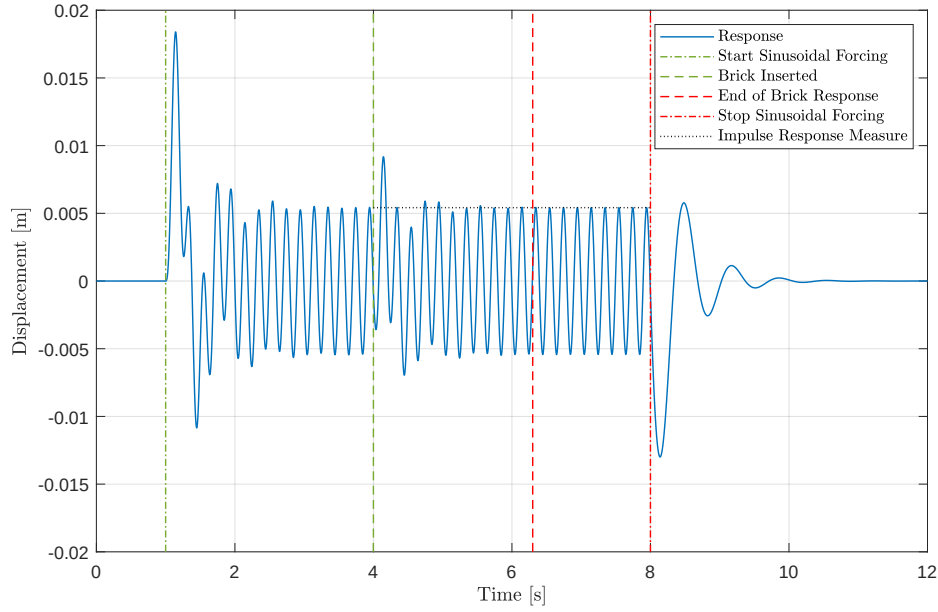


Figure 2: Washing machine displacement sequence.

The maximum displacement of the brick response is 9.2×10^{-3} m.

The steady-state displacement is 5.4×10^{-3} m, which agrees with 4.9×10^{-3} m found in Problem 1f.

Part 4c

The washing machine takes about 2.3 seconds to return to a steady-state response.

Part 4d

Figure 3 shows the velocity response of the washing machine.

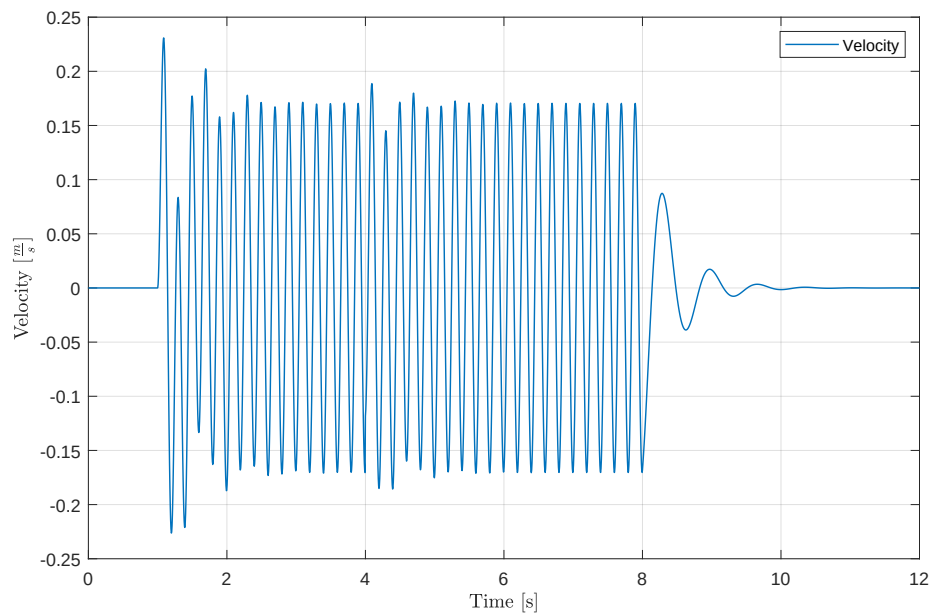


Figure 3: Washing machine velocity response.

Figure 4 shows the acceleration response of the washing machine.

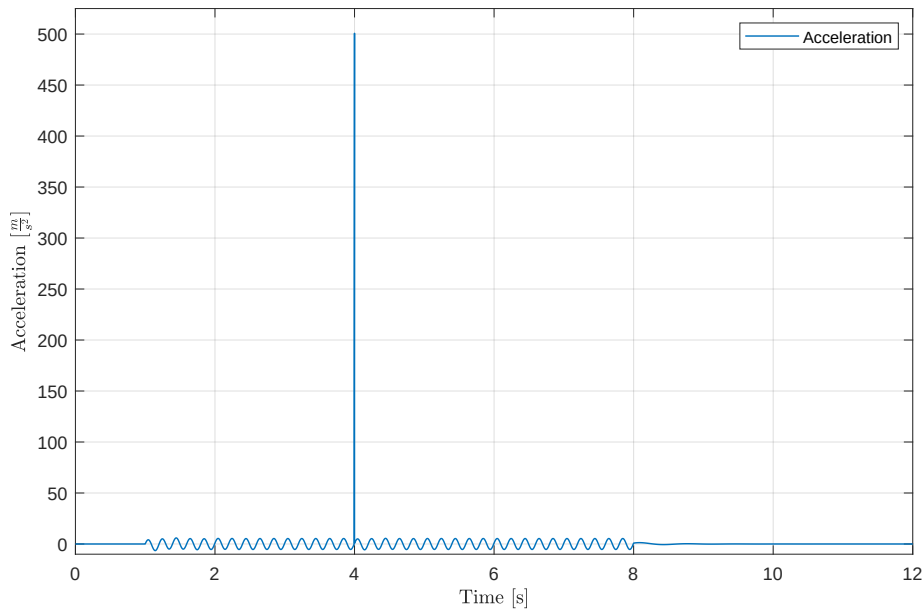


Figure 4: Washing machine acceleration response.

The maximum acceleration associated with the brick response is $500 \frac{\text{m}}{\text{s}^2}$.

Summary

Table 1 compares the admittance, displacement, and ground force for the three systems considered here.

	Admittance [$\frac{\text{m}}{\text{N}}$]	Displacement [m]	Ground Force [N]
1-DOF	1×10^{-5}	4.9×10^{-3}	4.8×10^1
2-DOF Optimized	9.22×10^{-6}	4.6×10^{-3}	4.6×10^{-5}
DVA	1.2×10^{-6}	3.8×10^{-6}	3.8×10^{-5}

Table 1: Performance comparison of the initial and optimized designs.

1 Appendix - Matlab Code for Problem 1

```
1
2
3
4 %% Synopsis
5
6 % Problem 1 – Washing Machine Mount Design – 1 Degree of Freedom (DOF)
7
8
9
10 %% Environment
11
12 close all; clear; clc;
13 % restoredefaultpath;
14
15 addpath( genpath( '../00 Support' ), '-begin' );
16
17 % set( 'groot', 'DefaultFigurePosition', [ 100 750 750 500 ] ); % x, y, width, height
18
19 set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
20 set( 0, 'DefaultFigureWindowStyle', 'normal' );
21 set( 0, 'DefaultLineLineWidth', 0.8 );
22 set( 0, 'DefaultTextInterpreter', 'Latex' );
23
24 format LongG;
25
26 pause( 1 );
27
28
29
30 %% Anonymous Function Definitions
31
32 h_force = @( force_mass, rotation_speed, load_distance ) force_mass .* rotation_speed.^2 .*
    load_distance;
33
34 h_transmissibility = @( epsilon, r ) sqrt( ( 1 + (2.*epsilon.*r).^2 ) ./ ( ( 1 - r.^2 ).^2 + (
    2.*epsilon.*r ).^2 ) );
35
36 rpm_conversion_to_radians_per_second = 0.10472;
37
38
39
40 %% Washing Machine Information
41
42 machine.mass = 100; % kg
43 machine.rotation_speed = 31.4; % radians\s or 300 rotations per minute (RPM)
44 machine.load_mass = 10; % kg
45 machine.static_displacement = 1e-2; % m
46
47
48
49 %% Load Force Information
50
51 dynamic_force.mass = 1; % kg
52 dynamic_force.radius = 0.5; % m
53
54 force_frequency = 300 * rpm_conversion_to_radians_per_second / ( 2 * pi ); % 5 Hz
55
56 % Maximum force applied to foundation is 100 N.
57
58
59
60 %% Part 1a
61
62 rpm = 0:1:350;
63 angular_velocity = rpm .* rpm_conversion_to_radians_per_second; % radians\s
64
65
66 figure( 'Name', 'Problem 1a – Load Force Versus Angular Velocity' ); ...
67 plot( rpm, h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius ) ); hold on;
68 plot( 300, 494, 'Marker', '.', 'MarkerSize', 20 );
69 line( [ 300, 300 ], [ 400 600 ], 'Color', 'r' );
70 text( 305, 494, '493.5 N' ); grid on;
71 xlabel( 'Angular Speed [RPM]' ); ylabel( 'Load Force [N]' );
72 axis( [ -10 355 -5 705 ] );
73
```

```

74
75
76 %% Part 1b
77
78 maximum_allowable_displacement = 1e-2; % m
79 ks = ( machine.load_mass * 9.81) / maximum_allowable_displacement; % 9,810 N/m
80
81
82 total_mass = machine.mass + machine.load_mass; % 110 kg
83 wo = sqrt( ks / total_mass ); % 9.4 radians/s
84 fo = wo / (2*pi); % 1.5 Hz = Natural frequency of the mount.
85
86
87
88 %% Part 1c
89
90 frequency_set = 0.1:0.01:1e3;
91 r = frequency_set ./ fo;
92
93 damping_ratio = logspace( log10(0.001), log10(1), 6 );
94
95
96 figure( 'Name', 'Problem 1c - Transmission Loss' ); ...
97 hold on;
98 %
99 for index = 1:numel( damping_ratio )
100     plot( r, 10*log10( h_transmissibility( damping_ratio( index ), r ) ) );
101 end
102 %
103 xlabel( 'Frequency Ratio,  $\frac{f}{f_0}$  [unitless]' ); ylabel( 'Transmission Loss [dB]' );
104 legend( '0.001', '0.004', '0.015', '0.063', '0.25', '1', 'Location', 'NorthEast' );
105 axis( [ 0.1 665 -60 25 ] );
106 grid on; box on;
107 set( gca, 'XScale', 'log' );
108
109
110
111 %% Part 1d
112
113 rpm = 0:1:350; % rotations per minute
114 rpm_conversion_to_radians_per_second = 0.10472; % radians/s
115 angular_velocity = rpm .* rpm_conversion_to_radians_per_second; % radians/s
116
117 frequency_set = angular_velocity / (2*pi);
118 r = frequency_set ./ fo;
119
120
121 figure( 'Name', 'Problem 1d - Applied Force to Foundation' ); ...
122 hold on;
123 %
124 for index = 1:numel( damping_ratio )
125     plot( rpm, h_transmissibility( damping_ratio( index ), r ) .* h_force( dynamic_force.mass
126         , angular_velocity, dynamic_force.radius ) );
127 %
128 line( [ 300, 300 ], [ 2 1e4 ], 'Color', 'k', 'LineStyle', '—' );
129 text( 305, 2, '300 RPM' ); grid on;
130 line( [ 0 350 ], [ 100 100 ], 'Color', 'r', 'LineStyle', '—' );
131 text( 0, 150, '100 N' ); grid on;
132 xlabel( 'Angular Speed [RPM]' ); ylabel( 'Force Applied to Foundation [N]' );
133 legend( '0.001', '0.004', '0.015', '0.063', '0.25', '1', 'Location', 'South' );
134 axis( [ -10 355 1 2e4 ] );
135 grid on; box on;
136 set( gca, 'YScale', 'log' );
137
138
139
140 %% Part 1e
141
142 % The best damping ratio is 0.25.
143
144 C = 0.25 * 2*sqrt( ks * total_mass ); % 519.4 kg/s
145
146 % The units of a viscous damping coefficient are Newton-seconds per meter (Ns/m) or kilograms per
147     second (kg/s).
148
149 m = total_mass;

```



```

150 k = [ ks 0 ];
151 dampings = [ C 0 ];
152 admittance = nDOF_direct_solution( m, k, dampings, frequency_set, 'admittance' );
153
154
155 figure( 'Name', 'Problem 1e - Admittance' ); ...
156 semilogy( rpm, abs( admittance ) ); grid on;
157 xlabel( 'Angular Speed [RPM]' ); ylabel( 'Admittance [ $\frac{m}{N}$ ]' );
158 set( gca, 'YScale', 'log' );
159 %
160 line( [ 300, 300 ], [ 7e-6 2e-5 ], 'Color', 'k', 'LineStyle', '—' );
161 text( 305, 2e-5, '300 RPM' ); grid on;
162 axis( [ -10 355 6e-6 2.5e-4 ] );
163
164
165 % temp2 = abs( admittance );
166 % round( temp2( 301 ), 3, 'significant' )
167
168
169


---


170 %% Part 1f
171
172 displacement = abs( admittance ) .* h_force( dynamic_force.mass, angular_velocity, dynamic_force.
    radius ).';
173
174
175 figure( 'Name', 'Displacement' ); ...
176 plot( rpm, displacement ); grid on;
177 xlabel( 'Angular Speed [RPM]' ); ylabel( 'Displacement [m]' );
178 line( [ 300, 300 ], [ 1e-3 1e-2 ], 'Color', 'k', 'LineStyle', '—' );
179 text( 305, 1e-3, '300 RPM' ); grid on;
180 set( gca, 'YScale', 'log' );
181 axis( [ -10 355 1e-7 2e-2 ] );
182
183
184 % temp2 = abs( displacement );
185 % round( temp2( 301 ), 3, 'significant' )
186
187
188


---


189 %% Clean-up
190
191 if ( ~isempty( findobj( 'Type', 'figure' ) ) )
192     monitors = get( 0, 'MonitorPositions' );
193     if ( size( monitors, 1 ) == 1 )
194         autoArrangeFigures( 3, 4, 1 );
195     elseif ( 1 < size( monitors, 1 ) )
196         autoArrangeFigures( 2, 2, 2 );
197     end
198 end
199
200 fprintf( 1, '\n\n*** Processing Complete ***\n\n' );
201
202
203


---


204 %% Reference(s)

```

2 Appendix - Matlab Code for Problem 2

```
1
2
3
4 %% Synopsis
5
6 % Problem 2 — Washing Machine Two-stage Mount Design
7
8
9
10 %% Note(s)
11
12 % A 2 degree-of-freedom (DOF) system will provide isolation performance 1 DOF system.
13
14 % Allowing the top and bottom masses, the washing machine and the raft ,
15 % respectively , to each be attached to ground independently.
16
17 % With the force applied to the washing machine and the displacement of the
18 % washing machine, use FSF( :, 1, 1 ). However, if the force is applied elsewhere ,
19 % than FSF( :, 2, 2, ) must be used.
20
21
22
23 %% Environment
24
25 close all; clear; clc;
26 % restoredefaultpath;
27
28 addpath( genpath( './00 Support' ), '-begin' );
29
30 % set( groot , 'DefaultFigurePosition', [ 100 750 750 500 ] ); % x, y, width, height
31
32 set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
33 set( 0, 'DefaultFigureWindowStyle', 'normal' );
34 set( 0, 'DefaultLineLineWidth', 0.8 );
35 set( 0, 'DefaultTextInterpreter', 'Latex' );
36
37 format ShortG;
38
39 pause( 1 );
40
41
42
43 %% Define Anonymous Functions
44
45 h_f_to_rpm = @( f ) ( f*2*pi ) / 0.10471;
46
47 h_w_to_rpm = @( w ) h_f_to_rpm( w*2*pi );
48 h_w_to_f = @( w ) w/(2*pi);
49
50 h_rpm_to_f = @( rpm ) ( rpm * 0.10471 ) / ( 2 * pi );
51
52
53 h_force = @( force_mass, rotation_speed, load_distance ) force_mass .* rotation_speed.^2 .*
    load_distance;
54
55
56
57 %% Problem 2a
58
59 % See report .
60
61
62
63 %% Problem 2b — Blocked Frequencies of Resonance
64
65 % The frequency of the forcing function is 5 Hz and the amplitude is 493.5 N.
66 fo = 5; % Hz
67 wo = 2*pi*fo; % 31.4 radians/s
68
69
70 switch ( 1 )
71
72     case 1 % Initial design parameters.
73
74         m1 = 100 + 10; % kg — Total mass of the washing machine and the load.
```

```

75         k1 = 1e2; % N\m
76         c1 = 5;
77
78         m2 = 100; % kg - UPPER LIMIT OF 100 kg
79         k2 = 1e2; % N\m
80         c2 = 5;
81
82         k3 = 1e3; % N\m -> Larger values produce stronger coupling (higher w+).
83
84
85     case 2 % Optimized design parameters.
86
87         m1 = 100 + 10; % kg - Total mass of the washing machine and the load.
88         k1 = 1e2; % N\m
89         c1 = 5;
90
91         m2 = 100; % kg - UPPER LIMIT OF 100 kg
92         k2 = 1e2; % N\m
93         c2 = 5;
94
95         k3 = 1e1; % N\m -> Larger values produce stronger coupling (higher w+).
96
97     end
98
99
100    w1 = sqrt( ( k1 + k3 ) / m1 ); % 3.2 radians\s; washing machine
101        f1 = w1 / (2*pi); % 0.5 Hz
102        rpm1 = f1*(2*pi)/0.10471; % 30.1 RPM
103
104    w2 = sqrt( ( k2 + k3 ) / m2 ); % 3.3 radians\s; raft
105        f2 = w2 / (2*pi); % 0.53 Hz
106        rpm2 = f2*(2*pi)/0.10471; % 31.7 RPM
107
108    % Note: If k3 is set to zero, then these frequencies represent the
109    % resonance frequencies for each mass on their own.
110
111
112


---


113 %% Problem 2b — Coupled Frequencies
114
115    mu_4 = ( k3^2 / (m1*m2) ); % 90.9 (radians\s)^4
116    mu = ( mu_4 ) ^0.25; % 3.1 radians\s
117
118    w(1) = 0.5*( ( w1^2 + w2^2 ) + sqrt( ( w1^2 - w2^2 )^2 + 4*mu_4 ) );
119    w(2) = 0.5*( ( w1^2 + w2^2 ) - sqrt( ( w1^2 - w2^2 )^2 + 4*mu_4 ) );
120    w = sqrt( w );
121
122
123    w_minus = w(2); % 0.97 radians\s (lower than w1, 1, and w2, 1.05 radians\s)
124    f_minus = w(2)/(2*pi); % 0.1545 Hz
125    rpm_minus = h_f_to_rpm( f_minus ); % 9.3 RPM
126    %
127    w_plus = w(1); % 4.5 radians\s (higher than w1, 1, and w2, 1.05 radians\s)
128    f_plus = w(1)/(2*pi); % 0.71 Hz
129    rpm_plus = h_f_to_rpm( f_plus ); % 42.8 RPM
130    %
131    % Note(s):
132    %
133    % The blocked frequencies, w1 and w2, are always between these two frequencies.
134
135    clear w mu_4;
136
137
138


---


139 %% Problem 2b — Mode Shapes
140
141    phi_plus = [ ...
142        1; ...
143        -(1/mu^2)*sqrt(m1/m2)*(w_plus^2 - w1^2), ...
144        ];
145    %
146    % 1
147    % -1.10
148
149    phi_minus = [ ...
150        1; ...
151        (1/mu^2)*sqrt(m1/m2)*(w1^2 - w_minus^2), ...
152        ];

```

```

153 %
154 % 1
155 % 0.99
156
157
158 % Note(s):
159 %
160 % When m1 = m2 and k1 = k3,
161 %
162 % a.) At w_minus, m1 and m2 move the same amount and are in-phase.
163 % b.) At w_plus, m1 and m2 move the same amount, but are out-of-phase.
164
165 % In general, for a 2 DOF system there will be 2 modes (in-phase and out-of-phase).
166
167 % With different initial conditions (no forcing), the behaviour is a weighted sum of the two
    modes.
168
169
170
171 %% Problem 2c
172
173 masses = [ m1 m2 ];
174 stiffnesses = [ k1 k3 k2 ];
175 dampings = [ c1 0 c2 ];
176
177
178 rpm = 0:0.1:500; % rotations per minute
179 rpm_conversion_to_radians_per_second = 0.10472;
180 angular_velocity = rpm .* rpm_conversion_to_radians_per_second; % radians\s
181 frequencies = angular_velocity / (2*pi); % Hz
182
183 FRF = nDOF_direct_solution( masses, stiffnesses, dampings, frequencies, 'admittance' );
184
185
186 admittance = zeros( numel( frequencies ), 1 );
187
188 for index = 1:1:numel( frequencies )
189     temp = diag( squeeze( FRF( index, 1, 1 ) ) ); % Check indexing to ensure force on washing
        machine.
190     admittance( index ) = abs( temp );
191 end
192 %
193 clear temp temp2;
194
195 washing_machine.admittance = admittance;
196
197
198 admittance = zeros( numel( frequencies ), 1 );
199
200 for index = 1:1:numel( frequencies )
201     temp = diag( squeeze( FRF( index, 1, 2 ) ) ); % Check indexing to ensure force on washing
        machine.
202     admittance( index ) = abs( temp );
203 end
204 %
205 clear temp temp2;
206
207 raft.admittance = admittance;
208
209
210 figure( 'Name', 'Admittance' ); ...
211 h1 = loglog( rpm, washing_machine.admittance, 'Color', 'r', 'LineWidth', 0.8 ); hold on;
212 h2 = loglog( rpm, raft.admittance, 'Color', 'b', 'LineWidth', 0.8 );
213 h3 = line( [ 300 300 ], [ 1e-6 1e-4 ], 'Color', 'k', 'LineStyle', '—' ); grid on;
214 text( 220, 2e-4, '300 RPM' );
215 %
216 h4 = line( [ rpm_minus rpm_minus ], [ 1e-8 1e1 ], 'Color', [ 0.72, 0.27, 1.00 ], 'LineStyle',
    '-.' );
217 %
218 h5 = line( [ rpm1 rpm1 ], [ 1e-8 1e1 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '-.' );
219 h6 = line( [ rpm2 rpm2 ], [ 1e-8 1e1 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '-' );
220 %
221 h7 = line( [ rpm_plus rpm_plus ], [ 1e-8 1e1 ], 'Color', [ 0.72, 0.27, 1.00 ], 'LineStyle',
    '-', '—' );
222 %
223 legend( [ h1 h2 h3 h4 h5 h6 h7 ], 'Washing Machine', 'Raft', 'Load Frequency', 'w-', 'w1', '
    w2', 'w+', 'Location', 'NorthEast' );
224 %

```

```

225     xlabel( 'Rotation [RPM]' ); ylabel( 'Admittance [ $\frac{m}{N}$ ]' );
226     axis( [ 6 400 1e-8 1e1 ] );
227
228
229 % round( washing_machine.admittance( 3001 ), 3, 'significant' )
230
231
232
233 %% Problem 2d
234
235 dynamic_force.mass = 1; % kg
236 dynamic_force.radius = 0.5; % m
237
238
239 figure( 'Name', 'Displacement' ); ...
240 h1 = loglog( rpm, washing_machine.admittance.*h_force( dynamic_force.mass, angular_velocity,
241     dynamic_force.radius ), 'Color', 'r' ); hold on;
242 h2 = loglog( rpm, raft.admittance.*h_force( dynamic_force.mass, angular_velocity,
243     dynamic_force.radius ), 'Color', 'b' );
244 h3 = line( [ 300 300 ], [ 1e-5 1e-1 ], 'Color', 'k', 'LineStyle', '—' ); grid on;
245 text( 220, 0.21, '300 RPM' );
246 legend( [ h1 h2 h3 ], 'Washing Machine', 'Raft', 'Load Frequency', 'Location', 'South' );
247 xlabel( 'Rotation [RPM]' ); ylabel( 'Displacement [m]' );
248 axis( [ 6 400 1e-6 1 ] );
249
250 % temp = washing_machine.admittance.*h_force( dynamic_force.mass, angular_velocity, dynamic_force
251     .radius );
252 % round( temp( 3001 ), 3, 'significant' )
253
254 % temp = raft.admittance.*h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius )
255     .';
256 % round( temp( 3001 ), 3, 'significant' )
257
258
259 %% Problem 2e
260
261 % Displacement of the raft multiplied by the K2 spring constant.
262
263 dynamic_force.mass = 1; % kg
264 dynamic_force.radius = 0.5; % m
265
266 h_force = @( force_mass, rotation_speed, load_distance ) force_mass .* rotation_speed.^2 .*
267     load_distance;
268
269 temp = h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius );
270
271 figure( 'Name', 'Force Applied to Ground by Raft' ); ...
272 h1 = loglog( rpm, raft.admittance.*temp*k2, 'Color', 'b' ); hold on;
273 h2 = line( [ 300 300 ], [ 2e-3 1e-1 ], 'Color', 'k', 'LineStyle', '—' ); grid on;
274 h3 = line( [ 6 400 ], [ 100 100 ], 'Color', 'r' );
275 text( 220, 2e-1, '300 RPM' );
276 legend( [ h1 h2 h3 ], 'Raft Force', 'Load Frequency', 'Maximum Floor Load', 'Location', '
277     South' );
278 xlabel( 'Rotation [RPM]' ); ylabel( 'Force Applied to Ground [N]' );
279 axis( [ 6 400 1e-3 1e3 ] );
280
281 % temp = raft.admittance.*h_force( dynamic_force.mass, angular_velocity, dynamic_force.radius )
282     .*k2;
283 % round( temp( 3001 ), 3, 'significant' )
284
285
286 %% Problem 2f
287
288 % See report.
289
290 %% Clean-up
291
292 if ( ~isempty( findobj( 'Type', 'figure' ) ) )
293     monitors = get( 0, 'MonitorPositions' );
294     if ( size( monitors, 1 ) == 1 )
295         autoArrangeFigures( 3, 4, 1 );

```

```

296         elseif ( 1 < size( monitors, 1 ) )
297             autoArrangeFigures( 2, 2, 2 );
298         end
299     end
300
301     fprintf( 1, '\n\n*** Processing Complete ***\n\n' );
302
303
304
305 %% Reference(s)
306
307
308
309 %% Comment(s)
310
311
312
313 %% Regions of Interest
314
315 % 5 regions of interest:
316
317 % 1.) Low frequency:  $w < w_{\text{minus}}$ :
318 %
319 % Stiffness controlled; two masses moving together; strong coupling ( $k_3$  high).
320
321
322 % 2.) Force at a resonance frequency:  $w = w_{\text{minus}}$  (lowest resonance frequency):
323 %
324 % At lowest resonance frequency; at  $w_{\text{minus}}$  mode, masses moving in same direction (in-phase)
    with same high amplitude
325
326
327 % 3.) Forcing frequency between  $w_{\text{minus}}$  and  $w_2$ 
328
329
330 % 4.) Forcing frequency  $w = w_2$ 
331
332
333 % 5.) Forcing frequency  $w_{\text{minus}} < w < w_+$ 
334 %
335 % No a special frequency.
336
337
338 % 6.) Forcing frequency  $w = w_+$ 
339 %
340 % At highest resonance frequency; at  $w_{\text{plus}}$  mode, masses moving in opposite directions (out-of-
    phase) with same high amplitude
341
342
343 % 7.) Forcing frequency  $w_+ < w$ 
344 %
345 % Mass controlled region; mass 1 (washing machine) moves more than mass 2 (raft);

```

3 Appendix - Matlab Code for Problem 3

```
1
2
3
4 %% Synopsis
5
6 % Problem 3 – Washing Machine Dynamic Vibration Absorber (DVA) Design
7
8
9
10 %% Note(s)
11
12 % At 300 RPM, the load frequency, this system will isolated vibration more
13 % than the 2 DOF and 1 DOF systems.
14
15 % The DVA behaves like a Helmholtz resonator, working as a mechanical notch
16 % filter at a particular frequency.
17
18 % DVAs work well in systems that have one dominate mode. A good
19 % example of this is building sway and its compensation. Based on the
20 % design and construction of a building, it will typically have a single,
21 % dominate mode.
22
23 % DVA is useful if you can not get away from the resonance frequency; can
24 % not tune a 2 DOF system when the resonance frequency is away from the
25 % forcing frequency.
26
27 % DVA introduces two resonances for the original resonance. These new
28 % resonances will produce greater motion at the new resonance frequencies.
29
30
31
32 %% Environment
33
34 close all; clear; clc;
35 % restoredefaultpath;
36
37 addpath( genpath( './00 Support' ), '-begin' );
38
39 % set( groot, 'DefaultFigurePosition', [ 100 750 750 500 ] ); % x, y, width, height
40
41 set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
42 set( 0, 'DefaultFigureWindowState', 'normal' );
43 set( 0, 'DefaultLineLineWidth', 0.8 );
44 set( 0, 'DefaultTextInterpreter', 'Latex' );
45
46 format ShortG;
47
48 pause( 1 );
49
50
51
52 %% Define Anonymous Functions
53
54 h_rpm_to_f = @( rpm ) ( rpm * 0.10471 ) / ( 2 * pi );
55 h_rpm_to_w = @( rpm ) ( rpm * 0.10471 );
56
57 h_f_to_rpm = @( f ) ( f*2*pi ) / 0.10471;
58
59 h_w_to_f = @( w ) w/(2*pi);
60 h_w_to_rpm = @( w ) h_f_to_rpm( w*2*pi );
61
62
63 h_force = @( force_mass, rotation_speed, load_distance ) force_mass .* rotation_speed.^2 .*
        load_distance;
64
65
66
67 %% Washing Machine Information
68
69 machine.mass = 100; % kg
70 machine.rotation_speed = 31.4; % radians\s or 300 rotations per minute (RPM)
71 machine.load_mass = 10; % kg
72 machine.static_displacement = 1e-2; % m
73
74
```

```

75
76 %% Load Force Information
77
78 dynamic_force.mass = 1; % kg
79 dynamic_force.radius = 0.5; % m
80
81 % Maximum force applied to foundation is 100 N.
82
83
84
85 %% Dynamic Vibration Absorber (DVA)
86
87 maximum_allowable_displacement = 1e-2; % m
88 ks = ( machine.load_mass * 9.81 ) / maximum_allowable_displacement; % 9,810 N\m
89
90
91 m = 110; % kg; washing machine mass and load mass
92 k = ks; % 9,810 N\m
93
94 wo = sqrt( k / m ); % 9.4 radians\s
95 fo = wo/(2*pi); % 1.5 Hz
96
97
98 m_dva = 15; % kg
99 k_dva = 1e5; % N\m
100
101 wo_dva = sqrt( k_dva / m_dva ); % 81.7 radians\s
102 fo_dva = wo_dva/(2*pi); % 13.0 Hz
103
104
105
106 %% Blocked Frequencies (Equation 9.1 of the Notes)
107
108 m1 = m_dva;
109 k1 = 0; % Spring above DVA mass; does not exist; set to zero.
110
111 m2 = m;
112 k2 = k;
113
114 k3 = k_dva;
115
116
117 w1 = sqrt( ( k1 + k3 ) / m1 ); % 81.7 radians\s; smaller than the system resonant frequency.
118 f1 = w1/(2*pi); % 13 Hz
119
120 w2 = sqrt( ( k2 + k3 ) / m2 ); % 31.6 radians\s; greater than the system resonant frequency.
121 f2 = w2/(2*pi); % 5.0 Hz
122
123
124
125 %% Coupled Frequencies (Equation 9.15 of the Notes)
126
127 mu_4 = ( k3^2 / (m1*m2) ); % 6.1 (kg\s)^4
128 mu = ( mu_4 )^0.25; % 49.1 kg\s
129
130 w(1) = 0.5*( ( w1^2 + w2^2 ) + sqrt( ( w1^2 - w2^2 )^2 + 4*mu_4 ) );
131 w(2) = 0.5*( ( w1^2 + w2^2 ) - sqrt( ( w1^2 - w2^2 )^2 + 4*mu_4 ) );
132 w = sqrt( w );
133
134
135 w_minus = w(2); % 8.9 radians\s (lower than 31.6 and 81.7 radians\s)
136 f_minus = w(2)/(2*pi); % 1.4 Hz
137 %
138 w_plus = w(1); % 87.1 radians\s (greater than 31.6 and 81.7 radians\s)
139 f_plus = w(1)/(2*pi); % 13.9 Hz
140
141
142 % The blocked frequencies are always between the lower and higher coupled frequencies.
143
144
145
146 %% Mode Shapes (Equations 9.18 and 9.19 of the Notes)
147
148 phi_plus = [ ...
149     1; ...
150     -(1/mu^2)*sqrt(m1/m2)*(w_plus^2 - w1^2), ...
151 ];
152 %

```



```

153 % 1
154 % -0.14
155
156
157 phi_minus = [ ...
158     1; ...
159     (1/mu^2)*sqrt(m1/m2)*(w1^2 - w_minus^2), ...
160 ];
161 %
162 % 1
163 % 0.99
164
165
166
167 %% Admittance Using Equations from DVA Lecture
168
169 MAXIMUM_RPM = 3e3;
170
171 F0 = 1;
172
173 w = 0:0.01:h_rpm_to_w( MAXIMUM_RPM );
174 f = w/(2*pi);
175 rpm = h_f_to_rpm( f );
176
177
178 m = [ 15 110 ];
179 k = [ 0 k_dva.*(1+0.1*1i) k2.*(1+0.2*1i) ];
180
181
182 admittance_modified = F0 ./ ( m(1)*m(2) ) * k(2) ./ (( w.^2 - w_plus^2 ) .* ( w.^2 - w_minus^2 ));
183 admittance_original = F0 ./ m(2) * 1 ./ ( w2^2 - w.^2 );
184
185
186 figure( 'Name', 'Admittance - Equations from DVA Lecture' ); ...
187 h1 = loglog( rpm, abs( admittance_original ) ); hold on;
188 h2 = loglog( rpm, abs( admittance_modified ) );
189 h3 = line( [ 300 300 ], [ 1e-6 1e-2 ], 'Color', 'k', 'LineStyle', '—' );
190 text( 220, 2e-2, '300 RPM' );
191 h4 = line( [ h_f_to_rpm( f_minus ) h_f_to_rpm( f_plus ) ], [ 1e-7 1e3 ], 'Color', [ 0.72,
192     0.27, 1.00 ], 'LineStyle', '—.' );
193
194 h5 = line( [ h_f_to_rpm( f1 ) h_f_to_rpm( f1 ) ], [ 1e-7 1e3 ], 'Color', [ 0.47, 0.67, 0.19
195     ], 'LineStyle', '—.' );
196 h6 = line( [ h_f_to_rpm( f2 ) h_f_to_rpm( f2 ) ], [ 1e-7 1e3 ], 'Color', [ 0.47, 0.67, 0.19
197     ], 'LineStyle', '—.' );
198
199 h7 = line( [ h_f_to_rpm( f_plus ) h_f_to_rpm( f_plus ) ], [ 1e-7 1e3 ], 'Color', [ 0.72,
200     0.27, 1.00 ], 'LineStyle', '—.' ); grid on;
201
202 axis( [ 5 3e3 1e-7 1 ] );
203 legend( [ h1 h2 h3 h4 h5 h6 h7 ], 'Original System', 'DVA System', 'Operating RPM', ...
204     '$w_-$', '$w_1$', ...
205     '$w_2$', '$w_+$', ...
206     'Location', 'NorthWest', 'Interpreter', 'Latex' );
207 xlabel( 'Rotation [RPM]' ); ylabel( 'Admittance [$\frac{m}{N}$]' );
208
209 %% Admittance Using nDOF Function
210
211 % Damping can be added by using a dampings vector or appending a complex
212 % value to the respective spring stiffness. THESE ARE NOT INTERCHANGABLE.
213
214 rpm = 0:0.1:MAXIMUM_RPM;
215 f = h_rpm_to_f( rpm );
216 w = f ./ (2*pi);
217
218 FRF = nDOF_direct_solution( m, k, [ 0 0 0 ], f, 'admittance' ); % Symmetric matrix.
219 %
220 %
221 % (1, 1) - Force on M1 and its associated displacement.
222 % (2, 2) - Force on M2 and its associated displacement.
223 %
224 % (1, 2) - Force on M1 and the displacement of M2.
225 % (2, 1) - Force on M2 and the displacement on M1.
226 %

```

```

227 % These are interchangeable; the same result.
228
229
230 figure( 'Name', 'Admittance - nDOF Calculated' ); ...
231 h1 = loglog( h_f_to_rpm( f ), abs( FRF( :, 1, 1 ) ), 'Color', 'r' ); hold on;
232 h2 = line( [ 300 300 ], [ 1e-6 1e-2 ], 'Color', 'k', 'LineStyle', '—' );
233 text( 220, 2e-2, '300 RPM' );
234 h3 = line( [ h_f_to_rpm( f_minus ) h_f_to_rpm( f_minus ) ], [ 1e-7 1e3 ], 'Color', [ 0.72,
0.27, 1.00 ], 'LineStyle', '—.' );
235
236 h4 = line( [ h_f_to_rpm( f1 ) h_f_to_rpm( f1 ) ], [ 1e-7 1e3 ], 'Color', [ 0.47, 0.67, 0.19
], 'LineStyle', '—.' );
237 h5 = line( [ h_f_to_rpm( f2 ) h_f_to_rpm( f2 ) ], [ 1e-7 1e3 ], 'Color', [ 0.47, 0.67, 0.19
], 'LineStyle', '—' );
238
239 h6 = line( [ h_f_to_rpm( f_plus ) h_f_to_rpm( f_plus ) ], [ 1e-7 1e3 ], 'Color', [ 0.72,
0.27, 1.00 ], 'LineStyle', '—' ); grid on;
240
241 axis( [ 2e1 MAXIMUM_RPM 1e-7 1 ] );
242 legend( [ h1 h2 h3 h4 h5 h6 ], 'DVA System', 'Operating RPM', ...
243 '$w_{-}$', '$w_{1}$', ...
244 '$w_{2}$', '$w_{+}$', ...
245 'Location', 'NorthWest', 'Interpreter', 'Latex' );
246 xlabel( 'Rotation [RPM]' ); ylabel( 'Admittance [ $\frac{m}{N}$ ]' );
247
248
249 % temp2 = abs( FRF( :, 1, 1 ) );
250 % round( temp2( 3001 ), 3, 'significant' )
251
252
253
254 %% Displacement
255
256 dynamic_force.mass = 1; % kg
257 dynamic_force.radius = 0.5; % m
258
259
260 figure( 'Name', 'Displacement' ); ...
261 h1 = loglog( rpm, abs( FRF( :, 1, 1 ) ).*h_force( dynamic_force.mass, w, dynamic_force.radius
).', 'Color', 'r' ); hold on;
262 h2 = line( [ 300 300 ], [ 1e-7 1e-3 ], 'Color', 'k', 'LineStyle', '—' ); grid on;
263 text( 220, 2e-3, '300 RPM' );
264 h4 = line( [ h_f_to_rpm( f_minus ) h_f_to_rpm( f_minus ) ], [ 1e-8 1e3 ], 'Color', [ 0.72,
0.27, 1.00 ], 'LineStyle', '—.' );
265
266 h5 = line( [ h_f_to_rpm( f1 ) h_f_to_rpm( f1 ) ], [ 1e-8 1e3 ], 'Color', [ 0.47, 0.67, 0.19
], 'LineStyle', '—.' );
267 h6 = line( [ h_f_to_rpm( f2 ) h_f_to_rpm( f2 ) ], [ 1e-8 1e3 ], 'Color', [ 0.47, 0.67, 0.19
], 'LineStyle', '—' );
268
269 h7 = line( [ h_f_to_rpm( f_plus ) h_f_to_rpm( f_plus ) ], [ 1e-8 1e3 ], 'Color', [ 0.72,
0.27, 1.00 ], 'LineStyle', '—' ); grid on;
270
271 legend( [ h1 h2 h3 h4 h5 h6 ], 'DVA System', 'Operating RPM', ...
272 '$w_{-}$', '$w_{1}$', ...
273 '$w_{2}$', '$w_{+}$', ...
274 'Location', 'NorthWest', 'Interpreter', 'Latex' );
275
276 xlabel( 'Rotation [RPM]' ); ylabel( 'Displacement [m]' );
277 axis( [ 2e1 MAXIMUM_RPM 1e-8 1 ] );
278
279
280 % temp2 = abs( FRF( :, 1, 1 ) ).*h_force( dynamic_force.mass, w, dynamic_force.radius ).';
281 % round( temp2( 3001 ), 3, 'significant' )
282
283
284
285 %% Ground Force
286
287 dynamic_force.mass = 1; % kg
288 dynamic_force.radius = 0.5; % m
289
290
291 figure( 'Name', 'Ground Force' ); ...
292 h1 = loglog( rpm, abs( FRF( :, 1, 1 ) ).*h_force( dynamic_force.mass, w, dynamic_force.radius
).'.*k2, 'Color', 'k' ); hold on;
293 h2 = line( [ 300 300 ], [ 1e-4 1e-0 ], 'Color', 'k', 'LineStyle', '—' ); grid on;
294 text( 220, 2e-0, '300 RPM' );

```

```

295     h3 = line( [ 6 MAXIMUM_RPM ], [ 100 100 ], 'Color', 'r' );
296     legend( [ h1 h2 h3 ], 'Ground Force', 'Operating RPM', 'Maximum Floor Load', 'Location',
297             'SouthEast' );
298     xlabel( 'Rotation [RPM]' ); ylabel( ' Force [N]' );
299     axis( [ 2e1 MAXIMUM_RPM 1e-5 1e3 ] );
300
301 % temp2 = abs( FRF( :, 1, 1 ) ).*h_force( dynamic_force.mass, w, dynamic_force.radius ).'*k2;
302 % round( temp2( 3001 ), 3, 'significant' )
303
304
305
306 %% Clean-up
307
308 if ( ~isempty( findobj( 'Type', 'figure' ) ) )
309     monitors = get( 0, 'MonitorPositions' );
310     if ( size( monitors, 1 ) == 1 )
311         autoArrangeFigures( 3, 4, 1 );
312     elseif ( 1 < size( monitors, 1 ) )
313         autoArrangeFigures( 2, 2, 2 );
314     end
315 end
316
317 fprintf( 1, '\n\n\n*** Processing Complete ***\n\n\n' );
318
319
320
321 %% Reference(s)

```

4 Appendix - Matlab Code for Problem 4

```
1
2
3
4 %% Synopsis
5
6 % Problem 4 – Washing Machine Abuse Testing
7
8
9
10 %% Environment
11
12 close all; clear; clc;
13 % restoredefaultpath;
14
15 addpath( genpath( './00 Support' ), '-begin' );
16
17 % set( 'groot', 'DefaultFigurePosition', [ 100 750 750 500 ] ); % x, y, width, height
18
19 set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
20 set( 0, 'DefaultFigureWindowStyle', 'normal' );
21 set( 0, 'DefaultLineLineWidth', 0.8 );
22 set( 0, 'DefaultTextInterpreter', 'Latex' );
23
24 format ShortG;
25
26 pause( 1 );
27
28
29
30 %% Define Anonymous Functions
31
32 h_unit_impulse = @( washing_machine_mass, wd, wo, epsilon, t ) ( 1./( washing_machine_mass.*wd)
33 ) .* exp( -wo.*epsilon.*t ) .* sin( wd.*t ); % mvo = 1 kg\ (m s)
34
35
36 %% Impulse Signal
37
38 time_step = 1e-4; % s
39 net_time = 10; % s
40 time_indices = 0:time_step:( net_time - time_step );
41
42 impulse = zeros( size( time_indices ) );
43 impulse( 4/time_step ) = 1./time_step;
44 brick_impulse = 5 .* impulse;
45
46
47
48 %% 1 DOF System Impulse Function
49
50 ks = 9810; % N\m
51 wo = sqrt( ks / 110 ); % 9.4 radians\s
52 fo = wo / (2*pi); % 1.5 Hz
53
54 epsilon = 0.25;
55 C = epsilon * 2*sqrt( ks * 110 ); % 519.4 kg\s
56
57 wd = wo * sqrt( 1 - epsilon^2 ); % 9.1 radians\s
58 fd = wd / (2*pi); % 1.46 Hz
59
60
61
62 %% 1 DOF System Impulse Response
63
64 impulse_response = h_unit_impulse( 100, wd, wo, epsilon, time_indices );
65
66 impulse_response_at_4 = conv( impulse, impulse_response ) * time_step;
67 time_indices_at_4 = ( 0:1:( numel( impulse_response_at_4 ) - 1 ) ) .* time_step;
68 %
69 figure( 'Name', '1 DOF System Impulse Response' ); ...
70 yyaxis left; ...
71 plot( time_indices, impulse ); grid on;
72 ylabel( 'Force [N]' );
73 ylim( 'auto' );
74 yyaxis right; ...
```

```

75     plot( time_indices_at_4, impulse_response_at_4 ); hold on;
76     ylabel( 'Admittance [ $\frac{m}{N}$ ]' );
77     ylim( 'auto' );
78     xlabel( 'Time [s]' );
79     xlim( [ 0 10 ] );
80
81
82
83 %% Combined Response
84
85 sinusoid_input = 493.5 * sin( 2 * pi * 5 * time_indices );
86 sinusoid_input( 1:1:( 1 / time_step ) ) = 0;
87 sinusoid_input( ( 8 / time_step ):1:( 10 / time_step ) ) = 0;
88
89
90 joint_signal = sinusoid_input + brick_impulse;
91
92
93 response_to_combined_forcing = conv( impulse_response, joint_signal ) .* time_step;
94 time_indices_sinusoidal_forcing = ( 0:1:( numel( response_to_combined_forcing ) - 1 ) ) .*
    time_step;
95 %
96 figure( 'Name', '1 DOF Combined Forcing Response' ); ...
97 h1 = plot( time_indices_sinusoidal_forcing, response_to_combined_forcing ); hold on;
98 %
99 h2 = line( [ 1 1 ], [ -2e-2 2e-2 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '-.' );
100
101 h3 = line( [ 4 4 ], [ -2e-2 2e-2 ], 'Color', [ 0.47, 0.67, 0.19 ], 'LineStyle', '—' );
102 h4 = line( [ 6.3 6.3 ], [ -2e-2 2e-2 ], 'Color', 'r', 'LineStyle', '—' );
103 %
104 h5 = line( [ 8 8 ], [ -2e-2 2e-2 ], 'Color', 'r', 'LineStyle', '-.' ); grid on;
105
106 h6 = line( [4 8 ], [ 5.42e-3 5.42e-3 ], 'LineStyle', ':', 'Color', 'k' );
107
108 legend( [ h1 h2 h3 h4 h5 h6 ], 'Response', 'Start Sinusoidal Forcing', ...
109     'Brick Inserted', 'End of Brick Response', ...
110     'Stop Sinusoidal Forcing', 'Impulse Response Measure', ...
111     'Location', 'NorthEast', 'Interpreter', 'Latex' );
112
113 xlabel( 'Time [s]' ); ylabel( 'Displacement [m]' );
114 axis( [ 0 12 -2e-2 2e-2 ] );
115
116
117
118 %% Acceleration
119
120 velocity = diff( response_to_combined_forcing ) ./ time_step;
121 %
122 figure( 'Name', 'Velocity' ); ...
123 plot( time_indices_sinusoidal_forcing(1:1:(end - 1)), velocity ); grid on;
124 legend( 'Velocity' );
125 xlabel( 'Time [s]' ); ylabel( 'Velocity [ $\frac{m}{s}$ ]' );
126 axis( [ 0 12 -0.25 0.25 ] );
127
128
129 acceleration = diff( velocity ) ./ time_step;
130 %
131 figure( 'Name', 'Acceleration' ); ...
132 plot( time_indices_sinusoidal_forcing(1:1:(end - 2)), acceleration ); grid on;
133 legend( 'Acceleration' );
134 xlabel( 'Time [s]' ); ylabel( 'Acceleration [ $\frac{m}{s^2}$ ]' );
135 axis( [ 0 12 -10 525 ] );
136
137
138
139 %% Clean-up
140
141 if ( ~isempty( findobj( 'Type', 'figure' ) ) )
142     monitors = get( 0, 'MonitorPositions' );
143     if ( size( monitors, 1 ) == 1 )
144         autoArrangeFigures( 3, 4, 1 );
145     elseif ( 1 < size( monitors, 1 ) )
146         autoArrangeFigures( 2, 2, 2 );
147     end
148 end
149
150 fprintf( 1, '\n\n*** Processing Complete ***\n\n' );
151

```

152
153
154 %% **Reference (s)**
